9701 - A Level Chemistry Definitions

Question	Answer
Lattice Energy	
Define lattice energy.	Energy change when 1 mole of an ionic compound is formed, from its gaseous ions.
Define enthalpy change of solution.	Enthalpy change when 1 mole of a solute dissolves in sufficient water to produce a very dilute solution.
Define enthalpy change of hydration.	Energy change when 1 mole of gaseous ions is dissolved in water.
Define standard enthalpy change of atomisation.	Enthalpy change when 1 mole of gaseous atoms is produced from its elements in its standard state under standard conditions.
Define first electron affinity.	Enthalpy change when 1 mole of electrons is gained by 1 mole of gaseous atoms to form 1 mole of gaseous -1 ions.
Suggest why the first electron affinity is generally exothermic whereas the second is endothermic.	Energy is now required to overcome the repulsion of the electron and the anion formed.
Electrochemistry	
Define standard electrode potential.	Voltage produced by a half cell, compared to a standard hydrogen electrode, under standard conditions
Define standard conditions.	1 mol dm^-3 solutions, 101 kPa pressure, 298K temperature.
State the function of a salt bridge.	Ions move to maintain the balance of charge to complete the circuit.
State the relationship between the Faraday constant, F, and the Avogadro constant, L.	F = Le
Further Equilibria	
Define conjugate acid-base pair.	Species that differ by a H+ ion.

Question	Answer
Define buffer solution.	Solution which resists changes in pH, when small amounts of acid or alkali are added.
Write an expression for Kw.	Kw = [H+][OH-].
Define partition coefficient.	Ratio of the concentrations of a solution dissolved in two different immiscible solvents, when an equilibrium has established at the boundary.
Reaction Kinetics	
Explain what is meant by order of reaction.	The power to which the concentration of a reactant is raised in the rate equation.
Explain what is meant by overall order of reaction.	The sum of the exponents to which the concentrations are raised to in the rate equation.
Explain what is meant by a heterogeneous catalyst.	A catalyst in a different state to the reactants.
Explain what is meant by a homogeneous catalyst.	A catalyst in the same state as the reactants.
Explain the term half-life of a reaction.	Time taken for the concentration of a reaction to fall to half of its original value.
Entropy and Gibbs Free Energy	
Define entropy.	Number of possible arrangements of particles/energy in a system.
Transition Elements	
Define transition element.	Element which forms one or more stable ions with incomplete d orbitals.
Define complex.	A central metal atom surrounded by one or more ligands.
Explain what is meant by coordination number.	The number of co-ordinate bonds being formed by the metal ion.
Explain why transition metals form complex ions.	Empty d orbitals are energetically accessible.
Define degenerate d orbital.	d orbitals are at same energy level.
Define non degenerate d orbitals.	d orbitals are at different energy levels.
Define monodentate ligand.	Species with one lone pair of electrons that forms a dative bond to a central metal

Question	Answer
	atom.
Explain what is meant by a bidentate ligand.	Species with two lone pairs of electrons that form dative bonds to a central metal atom.
Suggest the meaning of tridentate ligand.	Species with three lone pairs of electrons that form dative bonds to a central metal atom.
Explain what is meant by a polydentate ligand.	Species that donates more than two lone pairs of electrons that form dative bonds to a central metal atom.
Explain why transition metals behave as catalysts.	They have more than one stable oxidation state, and their d-orbitals are energetically accessible.
Explain why transition elements have variable oxidation states.	The 3d and 4s subshells are close in energy.
State two typical chemical properties of a transition element.	Variable oxidation states. Behave as catalysts. Form complex ions. Form coloured compounds.
Define stability constant.	Equilibrium constant for the formation of a complex in a solvent from its constituents.
Organic	
State what is meant by isoelectric point.	pH at which a molecule has no overall charge.
Synthesis and Analytical	
Explain the meaning of optically active.	Able to rotate the plane of polarised light.
Explain why CDCl3 is a better solvent than CHCl3 in proton NMR.	CDCl3 will not given an absorbtion peak.
Describe how the percentage composition of a mixture can be determined from a gas liquid chromatogram.	Area of peak divided by total area of all peaks x 100%
Describe how an Rf value can be calculated.	Rf = distance travelled by solute divided by distance travelled by solvent.
Describe what is meant by a racemic mixture.	A mixture containing equal amounts of each optical isomer.

Question	Answer
State the uses of TMS and D2O in NMR.	TMS = 0 ppm shift. D2O is used to identify O-H or N-H groups.
Define enantiomers.	Molecules that are non-superimposable mirror images, that rotate the plane of polarised light in the opposite direction.
State what is meant by retention time.	Time taken between injection and detection of substance.